

Choosing a Hypothesis Test

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).



- **Builds on:** Correlation ([Note 231](#)); Student's t-distribution ([Note 282](#)); Hypothesis testing: null and alternative ([Note 291](#)); Z-test and rejection regions ([Note 291](#)).
- **Compare with:** Chi-square tests ([Note 29](#)); One-way ANOVA ([Note 572](#)).

1. Overview

Key point: The kinds of features a question involves (categorical or numerical, one or two) decide which hypothesis test to run; every test then ends with the same rule, reject H_0 when $p \leq \alpha$.

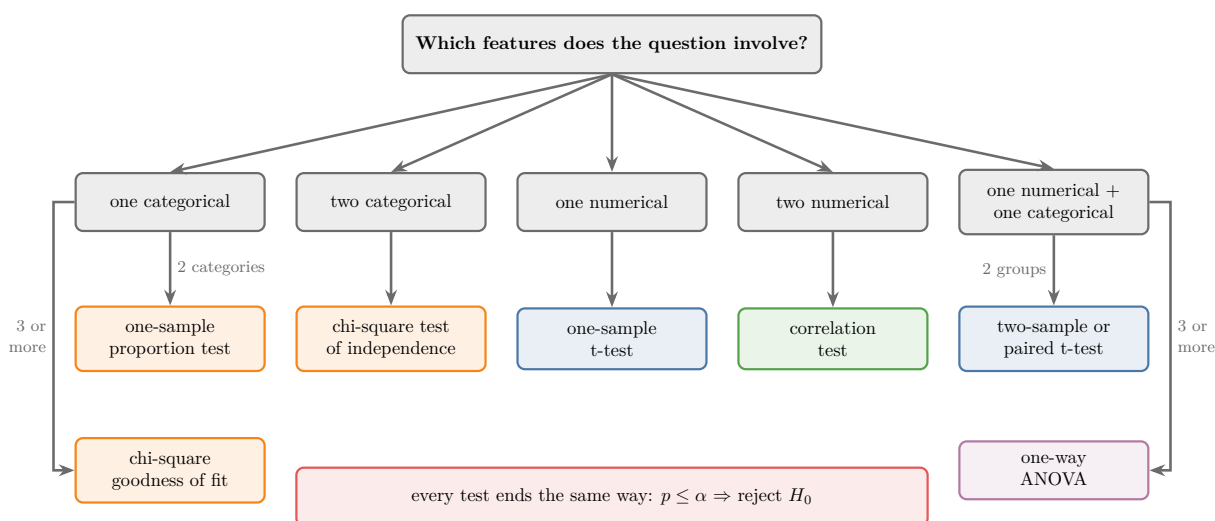


Figure 1: Choosing a test from the features involved

The earlier hypothesis-testing Notes built the z-test and three t-tests. Real questions come with data of many kinds: counts of men and women, heights, age groups. Figure 1 is the guide this Note follows: name the **features** the question is about, and the test follows. A feature is a variable of the data, one column of the data table, such as gender or height.

This Note covers:

- the decision logic shared by every test, with three common misreadings;
- a small dataset with two categorical and two numerical features;
- one test for each kind of question: the one-sample proportion test, the chi-square test, the t-tests, the correlation test and ANOVA.

Two of these tests are new here and taught in full: the one-sample proportion test and the correlation test. The chi-square tests and ANOVA each get their own Note: the [chi-square tests Note](#) and the [one-way ANOVA Note](#).

2. One logic for every test

Key point: State H_0 (“no difference”) and H_1 , fix α before looking, compute the p-value assuming H_0 , and reject H_0 when $p \leq \alpha$.

Every test in this Note follows the eight steps of a hypothesis test (see the [null and alternative hypotheses Note](#)). In short:

1. **Hypotheses.** H_0 says there is no difference or no relationship; H_1 says there is one.
2. **Significance level.** Fix α , usually 0.05, before running the test (see the [rejection region Note](#)).
3. **Test.** Choose the test from the features involved (Figure 1) and compute its statistic.
4. **P-value.** The probability, if H_0 were true, of a result at least as extreme as ours (see the [p-values Note](#)).
5. **Decision.** $p \leq \alpha$: reject H_0 . $p > \alpha$: fail to reject H_0 .

With $\alpha = 0.05$ in a two-tailed test, the rejection region is the outer 2.5% at each end of the test statistic’s distribution, 5% in total (see the [errors, power and tails Note](#)).

2.1 Three misreadings to avoid

Key point: H_0 is assumed, not known to be true; the p-value is not the probability that H_1 is true; and a large p-value does not let us “accept” H_0 .

These three mix-ups are common enough to correct here; each is explained in the Note linked.

- “ H_0 is always true.” We **assume** H_0 while computing the p-value, as a starting point. The test exists because H_0 may be false.
- “The p-value is the probability that H_1 is true.” The p-value is the probability of data this extreme if H_0 were true (see the [p-values Note](#)). The p-value says nothing directly about how likely either hypothesis is.
- “ $p > 0.05$, so we accept H_0 .” A large p-value means the sample gives too little evidence against H_0 . We say we **fail to reject** H_0 ; absence of evidence is not proof (see the [null and alternative hypotheses Note](#)).

3. The example dataset

Key point: 60 people with two categorical features (gender, age group) and two numerical features (height, weight); each pair of feature kinds leads to a different test.

We use one sample of 60 people with four features. Each person is one **observation** (one record, a row of the table):

gender	age_group	height (m)	weight (kg)
male	adult	1.67	71.0
male	adult	1.76	75.6
female	elderly	1.57	63.1
female	child	1.29	33.9
female	child	1.17	18.2

The data is simulated (`data/make_people.py` builds `data/people.csv` with a fixed seed), so that every test has clean, reproducible numbers. The dataset has:

- **gender:** a categorical feature with two categories, 34 female and 26 male;
- **age_group:** a categorical feature with three categories, 20 child, 26 adult, 14 elderly;
- **height** and **weight:** numerical (continuous) features.

The dataset is a **sample**. A bar chart of gender shows the proportions in these 60 people; the test tells whether they say anything about the population the sample came from (see the [what is statistics Note](#)).

4. One categorical feature: the one-sample proportion test

Key point: Does the share of one category differ from a claimed value? Compare the sample proportion with the claim, in standard errors.

4.1 The question

Key point: Is the proportion of men in the population different from 0.5?

With one categorical feature, the natural question is about proportions: is there a difference between the proportion of men and women? With two categories, the share of men π settles the question:

$$H_0 : \pi = 0.5, \quad H_1 : \pi \neq 0.5$$

Our sample has 26 men out of 60, a **sample proportion** $\hat{p} = 26/60 = 0.433$. The sample proportion is below 0.5, but a sample of 60 can easily miss by that much.

4.2 The z statistic for a proportion

Key point: $z = (\hat{p} - \pi_0) / \sqrt{\pi_0(1 - \pi_0)/n}$: the distance of the sample proportion from the claim, in standard errors.

A proportion is the mean of a 0/1 column, with standard deviation $\sqrt{\pi(1 - \pi)}$ (the Bernoulli variance, see the [Bernoulli and binomial Note](#)). By the central limit theorem the sample

proportion is close to normal for large n (see the [sampling distribution Note](#)). So the z-test of the [rejection region Note](#) applies, with these pieces:

1. **In words:** the gap between the sample proportion and the claimed proportion, divided by the standard error the claim implies.
2. **Formula:**

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\pi_0(1 - \pi_0)/n}}$$

3. **Example:** $\hat{p} = 0.433$, $\pi_0 = 0.5$, $n = 60$:

$$SE = \sqrt{\frac{0.5 \times 0.5}{60}} = \sqrt{0.00417} = 0.0645, \quad z = \frac{0.433 - 0.5}{0.0645} = \frac{-0.067}{0.0645} = -1.03$$

The standard error uses the claimed π_0 , not $\hat{p}$: the test asks how the sample would behave if H_0 were true.

4.3 The decision

Key point: $z = -1.03$ gives $p = 0.30 > 0.05$: we fail to reject H_0 ; the sample does not show that men and women occur in different proportions.

The test is two-tailed, so the p-value is both tails beyond $|z| = 1.03$:

$$p = 2 P(Z \geq 1.03) = 2 \times 0.151 = 0.30$$

Since $0.30 > 0.05$, we fail to reject H_0 . A 26-to-34 split in 60 people is well within what a 50/50 population produces by chance.

Python: The z-test by hand, and scipy's exact version.

```
import numpy as np
from scipy import stats

p_hat, pi0, n = 26 / 60, 0.5, 60
z = (p_hat - pi0) / np.sqrt(pi0 * (1 - pi0) / n)
2 * stats.norm.sf(abs(z))          # 0.30, z-test
stats.binomtest(26, n=60, p=0.5)    # p = 0.37, exact
```

Extra: The count of men is binomial, $B(60, \pi)$, so the p-value can also be computed exactly from binomial tail areas, as for the coin in the [p-values Note](#); `stats.binomtest` does this. The exact test gives 0.37 here. The z version is the normal approximation, good when $n\pi_0$ and $n(1 - \pi_0)$ are both at least 5 (NIST Handbook §7.2.4); here both are 30.

Extra: With three or more categories (child, adult, elderly against claimed shares), one proportion no longer describes the feature. The test for that case is the **chi-square goodness-of-fit test** (see the [chi-square tests Note](#)).

5. Two categorical features: the chi-square test

Key point: Does the proportion of men differ between age groups? Two categorical features are tested with the chi-square test of independence.

Adding a second categorical feature changes the question: is there a difference between the proportion of men and women **across age groups**? Equivalently: are gender and age group related?

- H_0 : gender and age group are independent (the share of men is the same in every age group);
- H_1 : they are related.

The counts go in a contingency table, as in the [contingency tables Note](#); pandas builds it with `pd.crosstab` (see the [bivariate analysis Note](#)):

	child	adult	elderly
female	12	12	10
male	8	14	4

The **chi-square test of independence** gives $\chi^2 = 2.50$ with $p = 0.29$. Since $0.29 > 0.05$, we fail to reject H_0 : the sample does not show that the gender mix changes with age group. How the statistic is built is the subject of the [chi-square tests Note](#).

6. One numerical feature: the one-sample t-test

Key point: Does the mean height differ from a known earlier value? One numerical feature against a claimed mean is the one-sample t-test.

Our sample's mean height is 1.533 m. An earlier sample from the same population had a mean of 1.55 m. Is there a difference?

$$H_0 : \mu = 1.55, \quad H_1 : \mu \neq 1.55$$

The question calls for the one-sample t-test of the [one-sample t-test Note](#): $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ with $n - 1$ degrees of freedom. With $s = 0.211$ and $n = 60$:

$$t = \frac{1.5327 - 1.55}{0.211/\sqrt{60}} = \frac{-0.0173}{0.0273} = -0.64, \quad df = 59, \quad p = 0.53$$

We fail to reject H_0 : the mean height is consistent with 1.55 m.

7. Two numerical features: the correlation test

Key point: Are height and weight related? Pearson's r measures the relationship; the correlation test asks whether r is far enough from 0 to rule out chance.

7.1 Correlation as a test

Key point: H_0 : the population correlation ρ is 0, no linear relationship; H_1 : $\rho \neq 0$.

Pearson's correlation coefficient r runs from -1 to $+1$; a value near 0 means no linear relationship (see the [covariance and correlation Note](#)). But r comes from a sample. Two unrelated features give a small non-zero r in a sample by chance, so we test:

$$H_0 : \rho = 0 \text{ (no relationship)}, \quad H_1 : \rho \neq 0$$

Here ρ (rho) is the correlation in the population, and r is its estimate from the sample.

7.2 The test statistic

Key point: $t = r\sqrt{n-2}/\sqrt{1-r^2}$, compared with a t-distribution with $n-2$ degrees of freedom.

Under H_0 the statistic below follows Student's t-distribution (see the [t-procedure Note](#)) with $n-2$ degrees of freedom. So the correlation test is itself a t-test.

1. **In words:** scale r by the square root of the sample size, and divide by the share of variation r leaves unexplained.
2. **Formula:**

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad df = n-2$$

3. **Example:** $r = 0.30$ from $n = 30$ pairs:

$$t = \frac{0.30 \times \sqrt{28}}{\sqrt{1-0.09}} = \frac{0.30 \times 5.29}{0.954} = 1.66, \quad df = 28, \quad p = 0.11$$

So $r = 0.30$ from 30 pairs is **not** significant at 5%. The same $r = 0.30$ from 100 pairs gives $t = 3.11$ and $p = 0.002$: significant. Sample size matters as much as the size of r .

7.3 Height and weight

Key point: $r = 0.98$ from 60 people gives $p < 0.001$: height and weight are strongly related.

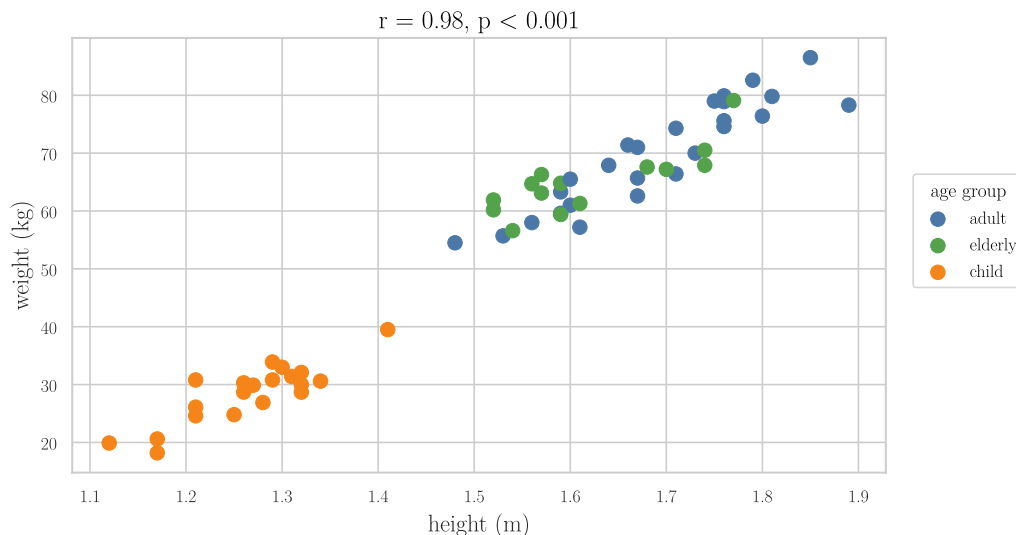


Figure 2: Height against weight for the 60 people

Figure 2 shows a tight upward pattern. Here $r = 0.980$ and $n = 60$:

$$t = \frac{0.980 \times \sqrt{58}}{\sqrt{1 - 0.961}} = 37.9, \quad p = 1.2 \times 10^{-42}$$

We reject H_0 : taller people in this population are heavier.

Python: `pearsonr` returns r and the p-value of the correlation test.

```
result = stats.pearsonr(people["height"], people["weight"])
result.statistic      #  $r = 0.980$ 
result.pvalue         #  $1.2e-42$ 
```

Extra: Part of this r comes from the gap between children and adults in Figure 2: two clouds far apart make a long line. The notebook tests this by dropping everyone but the adults: r falls from 0.98 to 0.92, lower but still strong. A significant correlation is also not causation (see the [covariance and correlation Note](#)).

8. One numerical and one categorical feature: t-test or ANOVA

Key point: A numerical feature compared across the groups of a categorical feature: two groups need a t-test, three or more need ANOVA.

8.1 Two groups: the two-sample t-test

Key point: Do adult men and adult women differ in mean height? Two independent groups: the two-sample t-test.

Gender has two categories, so comparing the height of adult men and women is the independent two-sample t-test (see the [two-sample and paired t-tests Note](#)). With 14 men (mean 1.758 m) and 12 women (mean 1.612 m):

$$t = 5.18, \quad df = 24, \quad p = 0.00003$$

We reject H_0 : adult men and women differ in mean height. If the two measurements came from the same subjects (weight before and after a diet), the paired t-test of the same Note applies instead.

8.2 Three or more groups: ANOVA

Key point: Does mean weight differ between children, adults and the elderly? Three groups: one-way ANOVA.

Age group has three categories. **One-way ANOVA** (analysis of variance) tests whether all three group means are equal:

$$H_0 : \mu_{\text{child}} = \mu_{\text{adult}} = \mu_{\text{elderly}}, \quad H_1 : \text{at least one mean differs}$$

The mean weights are 28.5, 69.8 and 65.0 kg. ANOVA gives $F = 203$ and $p < 0.001$, so we reject H_0 . The [one-way ANOVA Note](#) builds the F statistic and explains why three t-tests would not do.

Extra: With one numerical feature and **two** categorical features (weight by gender and age group together), the test is **two-way ANOVA**. Two-way ANOVA asks about each categorical feature and about their interaction (Montgomery, ch. 5).

9. Summary

Features in the question	Test	Result here
one categorical, 2 categories	one-sample proportion test	share of men: $z = -1.03$, $p = 0.30$
one categorical, 3+ categories	chi-square goodness of fit	see the chi-square tests Note
two categorical	chi-square test of independence	gender by age group: $\chi^2 = 2.50$, $p = 0.29$
one numerical	one-sample t-test	mean height: $t = -0.64$, $p = 0.53$
two numerical	correlation test	height and weight: $r = 0.98$, $p < 0.001$
numerical + 2 groups	two-sample t-test	adult height by gender: $p = 0.00003$
numerical + 3+ groups	one-way ANOVA	weight by age group: $F = 203$, $p < 0.001$

- The kinds of features decide the test; the decision rule is always $p \leq \alpha \Rightarrow \text{reject } H_0$.
- H_0 is assumed while computing p ; the p-value is not the probability of H_1 ; a large p means “fail to reject”, not “accept”.
- The one-sample proportion test is a z-test on $\hat{p}$ with standard error $\sqrt{\pi_0(1 - \pi_0)/n}$.
- The correlation test is a t-test on r with $n - 2$ degrees of freedom; a weak r needs a large sample to be significant.

10. Sources

- Montgomery, D. C. (2013). *Design and Analysis of Experiments*, 8th ed. Wiley. Chapter 5, factorial designs.
- NIST/SEMATECH (2012). *e-Handbook of Statistical Methods*. itl.nist.gov/div898/handbook. Section 7.2.4, testing a proportion (normal approximation when $\min(Np_0, N(1 - p_0)) \geq 5$).

11. Key terms

Term	Meaning
One-sample proportion test	A z-test of whether the proportion of one category in the population equals a claimed value π_0
Sample proportion $\hat{p}$	The share of a category in the sample, such as 26 men out of 60

Term	Meaning
Correlation test	A t-test of $H_0 : \rho = 0$, using $t = r\sqrt{n-2}/\sqrt{1-r^2}$ with $n-2$ degrees of freedom
Population correlation ρ	The correlation between two features in the whole population; r estimates it
Two-way ANOVA	ANOVA for one numerical feature and two categorical features